

III.2 Statistika kvantových plynů

→ grandkanonický model rovnovážného stavu

$$\hat{D}_G |_{\mathcal{H}_{Fock}^\pm} = \frac{1}{Z_G} e^{-\beta(\hat{H} - \mu\hat{N})}$$

partiční suma:

$$Z_G = \text{Tr}_{\mathcal{H}_{Fock}^\pm} e^{-\beta \sum_k (E_k - \mu) \hat{n}_k}$$

$$= \sum_{n_0, n_1, \dots} \underbrace{e^{-\beta \sum_k (E_k - \mu) n_k}}_{\prod_k e^{-\beta (E_k - \mu) n_k}} = \prod_k \sum_n \underbrace{(e^{-\beta (E_k - \mu)})^n}_{X_k}$$

$$\oplus \quad n \in \{0, \dots, \infty\} \Rightarrow \sum_n X_k^n = \frac{1}{1 - X_k}$$

$$\ominus \quad n \in \{0, 1\} \Rightarrow \sum_n X_k^n = 1 + X_k$$

$$= \prod_k (1 \mp X_k)^{\mp 1}$$

$$\ln Z_G^\pm = \mp \sum_k \ln [1 \mp e^{-\beta (E_k - \mu)}]$$

Statistické vlastnosti

Fermiony

$$\rightarrow \ln Z_G^- = \sum_k \ln (1 + e^{-\beta (E_k - \mu)})$$

$$\bullet \langle n_k \rangle = - \frac{\partial \ln Z_G^-}{\partial \lambda_k} = \frac{e^{-\lambda_k}}{1 + e^{-\lambda_k}} = \frac{1}{1 + e^{\beta (E_k - \mu)}} = f_k^-$$

$$\bullet \text{ pro fermiony } \langle n_k \rangle \equiv \mathbb{P}(n_k = 1)$$

$$\bullet \text{Cov}(n_k, n_l) = \frac{\partial^2 \ln Z_G^-}{\partial \lambda_k \partial \lambda_l} = -\delta_{kl} \frac{\partial \langle n_k \rangle}{\partial \lambda_k} = \delta_{kl} \frac{e^{\lambda_k}}{(1 + e^{\lambda_k})^2} \\ = f_k^- (1 - f_k^-) \delta_{kl}$$

Bosony

$$\rightarrow \ln Z_G^+ = - \sum_k \ln (1 - e^{-\lambda_k})$$

$$\bullet \langle n_k \rangle = - \frac{\partial \ln Z_G^+}{\partial \lambda_k} = \frac{e^{-\lambda_k}}{1 - e^{-\lambda_k}} = \frac{1}{e^{\beta (E_k - \mu)} - 1} = f_k^+$$

$$\bullet \text{Cov}(n_k, n_l) = \frac{\partial^2 \ln Z_G^+}{\partial \lambda_k \partial \lambda_l} = - \frac{e^{\lambda_k}}{(e^{\lambda_k} - 1)^2} \delta_{kl} \\ = f_k^+ (1 + f_k^+) \delta_{kl}$$

$$\langle n_k \rangle^\pm = f_k^\pm = \frac{1}{e^{\beta (E_k - \mu)} \mp 1}$$

$$\text{Cov}(n_k, n_l) = f_k^\pm (1 \pm f_k^\pm)$$

$$\mathbb{P}(n_k = n) = \begin{cases} f_k^- & \text{Bernoulliho rozdělení} \\ e^{-\lambda_k n_k} (1 - e^{-\lambda_k}) & \text{Geometrické rozdělení} \end{cases}$$